U.G.C. NET Ex
COMPUTE
(Solved with Expla

1. Which of the following is a sequential circuit?

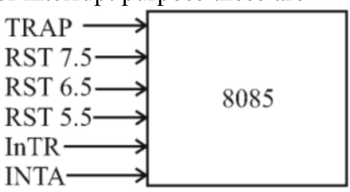
(a) Multiplexer (b) Decoder
(c) Counter (d) Full adder

Ans : (c) First 3 are combinational circuit (where output depends on present time)
Choice C is (counter) sequential circuit (sequential logic is a type of logic circuit whose output depends not only on the present value of its input but on the sequence of post inputs) and hence the answer is option (c).

2. 8085 microprocessor has .... hardware interrupts

(a) 2 (b) 3
(c) 4 (d) 5

Ans : (d) For interrupt purpose these are



TRAP
RST 7.5
RST 6.5
RST 5.5
InTR
INTA (It is not can interrupt pin but it is used to send acknowledgement of the interrupt request getting from other interrupt pin).
So answer is option (d).

3. Which of the following in 8085 microprocessor performs
HL = HL + DE ?

(a) DAD D (b) DAD H
(c) DAD B (d) DAD SP

Ans : (a) DAD instruction (double add) allows 16 bit addition between the HL register pair any, any one of the BC, DE, HL, SP register pairs
DAD D performs HL = HL + DE

4. The register that stores all interrupts requests is :

(a) Interrupt mask register
(b) Interrupt service register
(c) Interrupt request register
(d) Status register

Ans : (c) Interrupt request register (IRR) Indicates which interrupt request lines are active.

5. The .... addressing mode is similar to register indirect addressing mode, except that

an offset is added to the contents of the register. The offset and register are specified in the instruction.

(a) Base indexed
(b) Base indexed plus displacement
(c) Indexed
(d) Displacement

Ans : (d) The displacement addressing mode is similar to register indirect addressing mode, except that an offset is added to the contents of the register. The offset and register are specified in the instruction.

Based Indexed Addressing: The operand's offset is sum of the content of a base register BX or BP and an index register SI or DI.

Based Indexed plus displacement addressing mode: In this mode of addressing the operand's offset is given by offset=[BX or BP]+[SI or DI]+8 bit or 16 bit displacement. Indexed addressing mode: The operand's offset is the sum of the content of an index register SI or DI and an 8 bit or 16 bit displacement.

So, option (D) is correct.

6. In .... method, the word is written to the block in both the cache and main memory, in parallel.

(a) Write through (b) Write back
(c) Write protected (d) Direct mapping

Ans : (a) In write through method, data is written into the cache and the corresponding main memory location at the same time.

7. Which of the following statements concerning Object-Oriented databases is FALSE?

(a) Objects in an object-oriented database contain not only data but also methods for processing the data.
(b) Object-oriented databases store computational instructions in the same place as the data.
(c) Object-oriented databases are more adapt at handling structured (analytical) data than relational databases.
(d) Object-oriented databases store more types of data than relational databases and access that data faster.

Ans : (c) Objects in an object-oriented database contain not only data but also methods for processing the data. **Correct**.

Object-oriented databases store computational instructions in the same place as the data. **Correct**.

Object-oriented databases are more adapt at handling structured (analytical) data than relational databases. **Incorrect**
Object-oriented databases store more types of data than relational databases and access that data faster. **Correct**
So, option (C) is correct.

8. In distributed database, location transparency allows for database users, programmers and administrators to treat the data as if it is at one location. A SQL query with location transparency needs to specify :

(a) Inheritances (b) Fragments
(c) Locations (d) Local formats

Ans : (b) Fragmentation transparency is the highest level of transparency. The end user or programmer does not need to know that a database is partitioned. Therefore, neither prior to data access.

9. Consider the relations R(A, B) and S(B, C) and the following four relational algebra queries over R and S :

I. $\sqcap_{A.B}(R \bowtie S)$

II. $R \bowtie \sqcap_B(S)$

III. $R \cap (A(R) \times \sqcap_B(S))$

IV. $\sqcap_{A.R.B}(R \times S)$ **where R.B refers to the column B in table R.**

One can determine that :

(a) I, III and IV are the same query
(b) II, III and IV are the same query
(c) I, II and IV are the same query
(d) I, II and III are the same query

Ans : (d) Let us take an example as R = {(10, 2) (20, 2)}
S = {(2, 30); (3, 40)}
(I) R ⋈ S = {(20, 2, 30)} so πA, B R ⋈ S = {(20, 2)}
(II) πB(S) = {(2), (3)}, so R ⋈ π B (S) = {(20)2)}
(III) πA (R) × πB(S) = {10, 20} × {2, 3} ={(10, 2), (10, 3), (20, 2), (20, 3)} R∩πA(R) × πB(S) = {(20, 2)}
(IV) R × S = {(10, 1, 2, 30), (10, 1, 3, 40), (20, 2, 2, 30), (20, 2, 3, 40)} πA, R, B R × S = {(10, 2), (10, 3), (20, 2), (20, 3)}

10. Which of the following statements is TRUE?

D_1 : The decomposition of the schema R(A, B, C) into R_1(A, B) and R_2 (A, C) is always lossless.

D_2 : The decomposition of the schema R(A, B, C, D, E) having AD →B, C →DE, B → AE and AE →C, into R_1 (A, B, D) and R_2 (A, C, D, E) is lossless.

(a) Both D_1 and D_2 (b) Neither D_1 nor D_2
(c) Only D_1 (d) Only D_2

Ans : (d) Only D_2 Since AD is key and it present in both the tables not D_1
Since fd's not given
If we take B →A & C → A then it is lossy since no common attribute contain key from one of the table.

11. Consider the following ORACLE relations :
R (A, B, C) = {<1, 2, 3>, <1, 2, 0>, <1, 3, 1>, <6, 2, 3>, <1, 4, 2>, <3, 1, 4>}
S (B, C, D) + {<2, 3, 7>, <1, 4, 5>, <1, 2, 3>,<2, 3,4>, <3, 1, 4>}
Consider the following two SQL queries SQ_1 and SQ_2 :

SQ_1 :
```
SELECT R. B, AVG (S. B)
FROM R, S
WHERE R.A = S.C AND S.D < 7
GROUP BY R.B;
```
SQ_2 :
```
SELECT DISTINCT S.B, MIN (S.C)
FROM S
GROUP BY S.B
HAVING COUNT (DISTINCT S.D)>1;
```
If M is the number of tuples returned by SQ_1 and N is the number of tuples returned by SQ_2 then

(a) M = 4, N = 2
(b) M = 5, N = 3
(c) M = 2, N = 2
(d) M = 3, N = 3

Ans : (a)

Distinct S.B	Count(distinct S.d)
1	2
2	2
3	1

but the third tuple is not taken because count (distinct SD) >1

12. Semi-join strategies are techniques for query processing in distributed database system. Which of the following is a semi-join technique?

(a) Only the joining attributes are sent from one site to another and then all of the rows are returned
(b) All of the attributes are sent from one site to another and then only the required rows are returned
(c) Only the joining attributes are sent from site to another and then only the required rows are returned
(d) All of the attributes are sent from one site to another and then only the required rows are returned.

Ans : (c) Semi-join strategies are techniques for query processing in distributed database system. In which only the joining attributes are sent from one site to another and then only the required rows are returned. So, option (C) is correct.

13. Consider the Breshenham's circle generation algorithm for plotting a circle with centre (0, 0) and radius 'r' units in first quadrant. If the current point is (x_i, y_i) and decision parameter is pi then what will be the next point (x_{i+1}, y_{i+1}) and updated decision parameter p_{i+1} for $p_i \geq 0$?

(a) $x_{i+1} = x_i + 1$
$y_{i+1} = y_i$
$p_{i+1} = p_i + 4x_i + 6$

(b) $x_{i+1} = x_i + 1$
$y_{i+1} = y_i + 1$
$p_{i+1} = p_i + 4(x_i - y_i) + 10$
(c) $x_{i+1} = x_i$
$y_{i+1} = y_i - 1$
$p_{i+1} = p_i + 4(x_i - y_i) + 6$
(d) $x_{i+1} = x_i - 1$
$y_{i+1} = y_i$
$p_{i+1} = p_i + 4x_i + 10$

Ans : (a) **Bresenham's Algorithm**– We cannot display a continuous arc on the raster display. Instead, we have to choose the nearest pixel position to complete the arc.
From the following illustration, you can see that we have put the pixel at (X, Y) location and now need to decide where to put the next pixel − at N (X+1, Y) or at S (X+1, Y-1).
This can be decided by the decision parameter d.

- If d <= 0, then N(X+1, Y) is to be chosen as next pixel.
- If d > 0, then S(X+1, Y-1) is to be chosen as the next pixel.

14. A point P(5, 1) is rotated by 90° about a pivot point (2, 2). What is the coordinate of new transformed point P'?

(a) (3, 5) (b) (5, 3)
(c) (2, 4) (d) (1, 5)

Ans : (a) Transformation matrix for rotation to about a pivot point (2, 2) is given by
Ro, P = T –v Ro Tv where v = hitkJ

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ -h & -k & 1 \end{pmatrix} \begin{pmatrix} \cos 0 & \sin 0 & 0 \\ -\sin 0 & \cos 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ h & k & 1 \end{pmatrix}$$

$$= \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 2 & 2 & 1 \end{pmatrix} = \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ N & 0 & 1 \end{pmatrix}$$

Now P (S, 1) is rotated to P'

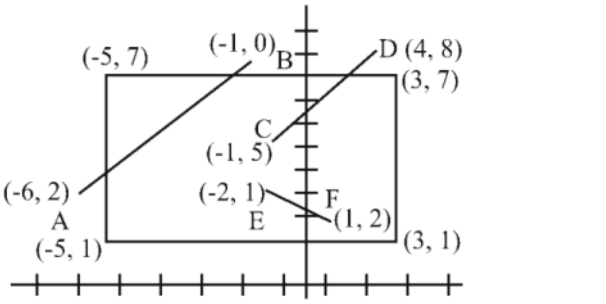
$$P' = [511] \begin{pmatrix} 0 & 1 & 0 \\ -1 & 0 & 0 \\ 4 & 0 & 1 \end{pmatrix} = [3,5,1]$$

So P' (x', y') = 3, 5 hence answer is DP (1)

15. Let R be the rectangular window which the lines are to be clipped using 2D Sutherland-Cohen line clipping algorithm. The rectangular window has lower left-hand corner at (–5, 1) and upper right-hand corner at (3, 7). Consider the following three lines for clipping with the given end point co-ordinates :
Line AB : A (–6, 2) and B (–1, 8)
Line CD : C (–1, 5) and D (4, 8)
Line EF : E (–2, 3) and F (1, 2)
Which of the following lines (s) is/are candidate for clipping?

(a) AB (b) CD
(c) EF (d) AB and CD

Ans : (d)



Clearly only AB & CD are candidate for line clipping.

16. In perspective projection, if a line segment joining a point which line in front of the viewer to a point in back of the viewer is projected to a broken line of infinite extent. This is known as ....

(a) View confusion
(b) Vanishing point
(c) Topological distortion
(d) Perspective foreshortening

Ans : (c) Topological distortion : Consider the plane that passes through the center of projection & parallel to view plane. The point of this plane are projected to infinity the perspective transformation.

17. Let us consider that the original point is (x, y) and new transformed point is (x', y') Further, Shx and Shy are shearing factors in x and y directions. If we perform the y-direction shear relative to x = xref then the transformed point is given by ....

(a) $x' = x + Sh_x . (y - y_{ref})$
$y' = y$
(b) $x' = x$
$y' = y . Sh_x$
(c) $x' = x$
$y' = Sh_y (x - x_{ref}) + y$
(d) $x' = Sh_y . y$
$y' = y . (x - x_{ref})$

Ans : (c) Ans is (C) if we perform x direction shear with reference to y axis, then x′= x + Shx.(y−yref); y′=y if we perform y direction shear with reference to x axis, then
x′ = x; y′ = Shy(x−xref) + y

18. Which of the following statement (s) is/are correct with reference to curve generation?

I. Hermite curves are generated using the concepts of interpolation.
II. Bezier curves are generated using the concepts of approximation.
III. The Bazier curve lies entirely within the convex hull of its control points.
IV. The degree of Bezier curve does not depend on the number of control points.

(a) I, II and IV only (b) II and III only
(c) I and II only (d) I, II and III only

Ans : (d) Hermits curves are very easy to calculate but also very powerful. They are used to smoothly interpolate between key points.
Bezier curve section can be fitted to any number of control points. The number of control points. The number of control points to be approximated & their relative position determine the degree of the Bezier polynomial.
So answer is option (d).

19. Given the following statements :

(1) To implement Abstract data type, a programming language require a syntactic unit to encapsulate type definition.
(2) To implement ADT, a programming language requires some primitive operations that are built in the language processor.
(3) C++, Ada, Java 5.0, C#2005 provide support for parameterized ADT

Which one of the following options is correct?

(a) (1), (2) and (3) are false
(b) (1) and (2) are true; (3) is false
(c) (1) is true; (2) and (3) are false
(d) (1), (2) and (3) are true

Ans : (d) Language requirements for ADT'S
- A syntactic unit in which to encapsulate the type definition
- Ada provides package that simulate ADTs
- C++ data abstraction is provided by classes
- Java's data abstraction is similar to C++

so A, B, C are true
Hence answer is (d)

20. Match the following types of variables with the corresponding programming language :

A.	**Static variables**	**i.**	**Local variables in Pascal**
B.	**Stack dynamic**	**ii.**	**All variables in APL**
C.	**Explicit heap dynamic**	**iii.**	**Fortran 77**
D.	**implicit heap dynamic**	**iv.**	**All objects in JAVA**

Code :

	A	B	C	D
(a)	(i)	(iii)	(iv)	(ii)
(b)	(iv)	(i)	(iii)	(ii)
(c)	(iii)	(i)	(iv)	(ii)
(d)	(ii)	(i)	(iii)	(iv)

Ans : (c) FORTRAN I, II & IV contain only static variables. All objects in Java are explicit heap dynamic variable & are destroyed by "Garbage Collector" Languages Such as APL, ALGOL 68, & LISP Use implicit heap dynamic variables.
So, a – 111, C – iv, d –11
Hence correct option is (c)

21. Which of the following if false regarding the evaluation of computer programming language?

(a) Application oriented features
(b) Efficiency and Readability
(c) Software development
(d) Hardware maintenance cost

Ans : (d) Programming language evaluation is no way related with cost of H/w, so answer is option (d).

22. The symmetric difference of two sets S1 and S2 is defined as
$S_1 \Theta S_2 = \{x | x \in S_1$ or $x \in S_2$, but x is not both S_1 and $S_2\}$
The nor of two language is defined as
nor $(L_1, L_2) = \{w | w \in L_1$ and $w \not\in L_2\}$
Which of the following is correct?

(a) The family of regular language is closed under symmetric difference but not closed under nor.
(b) The family of regular languages is closed under nor but not closed under symmetric difference.
(c) The family of regular languages are closed under both symmetric difference and nor.
(d) The family of regular language are not under both symmetric difference and nor.

Ans : (c) S1 ⊕ S2 = 'S2 + S1S2'
Regular language are closed union, intersection and hence they are closed under set difference.
nor $(L_1 + L_2) = L_1 . L_2$ (from demorgan law)
here represent intersection.
Regular language are closed under complementation & intersection and hence they are closed under nor.

23. The regular expression for the complement of the language $L = \{a^n b^m | n \geq 4, m \leq 3\}$ is :

(a) $(\lambda + a + aa + aaa)$ b* + a* bbbb* + (a + b)* ba (a + b)*
(b) $(\lambda + a + aa + aaa)$ b* + a* bbbbb* + (a + b)* ab (a + b)*
(c) $(\lambda + a + aa + aaa)$ + a* bbbbb* + (a + b)* ab(a + b)*
(d) $(\lambda + a + aa + aaa)$b* + a* bbbbb* + (a + b)* ba(a + b)*

Ans : (d)
(1) Using the term a*bbbb* it can accept a^4b^3 hence it is not complement
(2) Using (a + b) * ab (a + b)* it can accept + a^4b^3 hence it is not complement
(3) Using (a + b)* ab (a + b)* it can accept a^4b^3 hence it is not complement
(4) (a+b)*ba (a + b)* cannot produce strings a^nb^m format & other two terms cannot produce any strings a^nb^m such that n > = 4 & m < = 3

24. Consider the following two languages :

$L_1 = \{0^i1^j | \gcd(i, j) = 1\}$

L_2 is any subset os 0*

Which of the following is correct?

(a) L_1 is regular and L_2* is not regular

(b) L_1 is not regular and L_2* is regular

(c) Both L_1 and L_2* are regular languages

(d) Both L_1 and L_2* are not regular languages

Ans : (b) If L is regular then of course L* is also regular. If L is infinite, then it is regular and again L* is regular for L = { 0p p is prime}, L is not regular, L ≤ {0} * & L * is regular

25. If link transmits 4000 frames per second and each slot has 8 bits, the transmission rate of circuit of this TDM is ....

(a) 64 Kbps (b) 32 MbpS

(c) 32 Kbps (d) 64 MbpS

Ans : (c) Transmission rate = frame rate * no of bits in a slot. Frame rate = 4000 frames per second # of bits in a slot = 8 bit TDM = 4000 * 8 bps = 32 Kbps. So, option (C) is correct.

26. Given the following statements :

(1) Frequency division multiplexing is a technique that can be applied when the bandwidth of a link is greater than combined bandwidth of signals to be transmitted.

(2) Wavelength division multiplexing (WDM) is an analog multiplexing technique to combine optical signals.

(3) WDM is a digital multiplexing technique.

(4) TDM is digital multiplexing technique.

Which of the following is correct?

(a) (1), (2), (3) and (4) are true

(b) (1), (2), (3) and (4) are false

(c) (1), (2) and (4) are false (3) is true

(d) (1), (2) and (4) are true (3) is false

Ans : (d) Multiplexing can be of three types

(1) Frequency division multiplexing (FDM) : analog signal

(2) wave division multiplexing (WDM) : analog signal (used in optical fiber)

(3) Time division multiplexing (TDM) : digital signals : : In TDM, a single frame is made with interleaved digital signals from n devices. In TDM, the transmission rate of the multiplexed path is greater than the sum of transmission rates of all sources.

27. A pure ALOHA network transmits 200 bit frames using a shared channel with 200 Kbps bandwidth. If the system (all stations put together) produces 500 frames per second, then the throughput of the system is ....

(a) 0.384 (b) 0.184

(c) 0.286 (d) 0.586

Ans : (b) Throughput for pure ALOHA is $s = G \times e^{-2}$, where G is the load factor and take e = 2.71
The throughput is maximum, when G = 1/2 which is S_{max} = 0.184 Transmission time of rate = 200 bits/200 kbps = 1 ms
Now, system as taken all together produce 500 frames per seci ; e
Now, as load factor G = 1/2, then throughput is maximum for pure aloha
So answer is (b)

28. Match the following :

A.	**Line coding**	**i.**	**A technique to change analog signal to digital data**
B.	**Block coding**	**ii.**	**Provides synchronization without increasing number of bits**
C.	**Scrambling**	**iii.**	**Process of converting digital data to digital signal**
D.	**Pulse code modulation**	**iv.**	**Provides redundancy to ensure synchronization and inherits error detection**

Code :

	A	B	C	D
(a)	(iv)	(iii)	(ii)	(i)
(b)	(iii)	(iv)	(ii)	(i)
(c)	(i)	(iii)	(ii)	(iv)
(d)	(ii)	(i)	(iv)	(iii)

Ans : (b)
(a) Line coding is a process for converting digital data into digital signal (III)
(b) Block coding is the way the original number of bits is increased
(c) Scrambling is a technique that substitutes long zero level poises with a combination of other levels without increasing the number of bits (II)
(d) Pulse code modulation (PCM) is a technique to change analog signal to digital data (I)

29. Assume that we need to download text documents at the rate of 100 pages per minute. A page is an average of 24 lines with 80 characters in each line and each character requires 8 bits. Then the required bit rate of the channel is ....

(a) 192 mbps (b) 512 mbps
(c) 1.248 mbps (d) 1.536 mbps

Ans : (d) downloading rate = 100 page
$\Rightarrow$ $100 \times 24 \times 80 \times 8 = 1536000$ bit rate per second
$\Rightarrow$ 1.536 mbps

30. Encrypt the plain text message "EXTRANET" using transposition cipher technique with the following key :

3	**5**	**2**	**1**	**4**	**(Cipher text)**
1	**2**	**3**	**4**	**5**	**(Plan text)**

Using 'Z' as bogus character.

(a) TEXERTZENZ (b) EXTRANETZZ
(c) EZXZTRZANZET (d) EXTZRANZETZ

Ans : (a)Plain text

1	2	3	4	5	1	2	3	4	5
E	X	T	R	A	N	E	T	Z	Z

Cipher text

3	5	2	1	4	3	5	2	1	4
T	E	X	E	R	T	Z	E	N	Z

31. The number of different binary trees with 6 nodes is ....

(a) 6 (b) 42
(c) 132 (d) 256

Ans : (c) (2n)!/(n + 1)!n! different binary trees are possible with n given n = 6
number of trees = 12!/(7 : × 6 :) = 132

32. Let A[1 ......n] be an array of n distinct number. if i < j and A[i] > A[j], then the pair (i, j) is called an inversion of A. What is the expected number of inversions in any permutation on n elements?

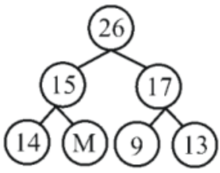
(a) $\theta(n)$ (b) $\theta(\lg n)$
(c) $\theta(n\lg n)$ (d) $\theta(n^2)$

Ans : (d) Answer is option (d) Expected number of inversion = n(n –s)/2 for randomly $\Rightarrow$ n (n – 1)/4

33. Which one of the following array represents s binary max-heap?

(a) [26, 13, 17, 14, 11, 9, 15]
(b) [26, 15, 14, 17, 11, 9, 13]
(c) [26, 15, 17, 14, 11, 9, 13]
(d) [26, 15, 13, 14, 11, 9, 17]

Ans : (c) Rest option did not satisfy max heap property, so ans. is (c)



34. Match the following :

A.	**Huffman codes**	**i.**	**$O(n^2)$**
B.	**Optimal polygon triangulation**	**ii.**	**$\theta(n^3)$**
C.	**Activity selection problem**	**iii.**	**O(nlgn)**
D.	**Quicksort**	**iv.**	**$\theta(n)$**

Code :

	A	B	C	D
(a)	(i)	(ii)	(iv)	(iii)
(b)	(i)	(iv)	(ii)	(iii)
(c)	(iii)	(ii)	(iv)	(i)
(d)	(iii)	(iv)	(ii)	(i)

Ans : (c)

35. Suppose that we have numbers between 1 and 1,000 in a binary search tree and want to search for the number 364. Which of the following sequences could not be the sequence of nodes examined?

(a) 925, 221, 912, 245, 899, 259, 363, 364
(b) 3, 400, 388, 220, 267, 383, 382, 279, 364
(c) 926, 203, 912, 241, 913, 246, 364
(d) 3, 253, 402, 399, 331, 345, 398, 364

Ans : (c) g13 can not be searched after 912, after 912 any number must be less than 912 since 912 > 364 So we will move towards 912 to 1, so ans. is (c)

36. A triangulation of a polygon is a set of T chords that divide the polygon into disjoint triangles. Every triangulation of n-vertex convex polygon has .... chords and divides the polygon into ...... triangles.

(a) n – 2, n – 1 (b) n – 3, n – 2
(c) n – 1, n (d) n – 2, n – 2

Ans : (b) A triangulation of a polygon is a set of T chords that divide the polygon into disjoint triangles. Every triangulation of n-vertex convex polygon has n – 3chords and divides the polygon into n – 2triangles.

37. Implicit return type of a class constructor is :

(a) not of class type itself
(b) class type itself
(c) a destructor of class type
(d) a destructor not of class type

Ans : (b) The implicit return type of a class constructor is the class type i-self. It is the constructors job to initials the internal state of an object so that the code creating an instance will have a fully initialized, usable object immediately.

38. It is possible to define a class within a class termed as nested class. There are .... types of nested classes.

(a) 2 (b) 3
(c) 4 (d) 5

Ans : (c) There are four kinds of nested class in Java.

- Static class : declared as a static member of another class
- Inner class : declared as a instance member of another class
- Local interclass : declared inside an instance method of another class.
- anonymous inner class : Like a local inner class, but written as an expression which returns a one off object.

39. Which of the following statements is correct?

(a) Aggregation is a strong type of association between two classes with full ownership.
(b) Aggregation is a strong type of association between two classes with partial ownership.
(c) Aggregation is a weak type of association between two classes with partial.
(d) Aggregation is a weak type of association between two classes with full ownership.

Ans : (c) Aggregation is a weak type of association with partial ownership.

40. Which of the following statements is correct?

(a) Every class containing abstract method must not be declared abstract.
(b) Abstract class cannot be directly initiated with 'new' operator.
(c) Abstract class cannot be initiated.
(d) Abstract class contains definition of implementation.

Ans : (c) Abstract classes are classes that contain one or more abstract method.

- Abstract method is method that is declared, but contains no implementation
- Abstract classes may not be instantiated & require subclass to provide implementation for the

so answer is op. (c)

41. Which of the following statements is not correct?

(a) HTML is not screen precise formatting language
(b) HTML does not specify a logic
(c) DHTML is used for developing highly interactive web pages
(d) HTML is a programming language

Ans : (d) HTML is not programming language programming language means where writing logic is possible which is not supported by HTML.

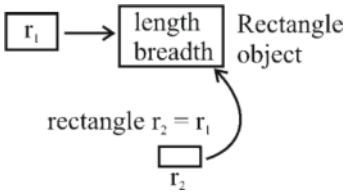
42. When one object reference variable is assigned to another object reference variable then

(a) a copy of the object is created
(b) a copy of the reference is created
(c) a copy of the reference is not created
(d) it is illegal to assign one object reference variable to another object reference variable

Ans : (b)

(1) We can assign value of reference variable to another reference variable.
(2) Reference variable is used to store the address of the variable
(3) Assigning Reference will not create distinct copies of object
(4) All reference variables are referring to some object
(5) Assigning object reference variable does not create distinct object

rectangle r1 = now rectangle ( ) :



43. A server crashes on the average once in 30 days, that is the Mean Time Between Failures (MTBF) is 30 days. When this happens, it takes 12 hours to reboot it, that is, the Mean Time to Repair (MTTR) is 12 hours. The availability of server with these reliability data values is approximately :

(a) 96.3% (b) 97.3%
(c) 98.3% (d) 99.3%

Ans : (c) If we let A represent availability, then the simplest formula for availability
A = uptime/(uptime + downtime)
MTBF is mean time between failures
A = MTTR is mean time to repair
A = MTBF/(MTBF + MTTR)
So MTBF = 30 days = 30 × 24 = 720 hr
A = 720/(720 + 12)
= 98.3%

44. Match the software maintenance activities in List-I to its meaning in List-II

List-I		**List-II**	
i.	**Corrective**	**A.**	**Concerned with performing activities to reduce the software complexity thereby improving program understandability and increasing software maintainability**
ii.	**Adaptive**	**B.**	**Concerned with fixing errors that are observed when the software is in use.**
iii.	**Perfective**	**C.**	**Concerned with the change in the software that takes place to make the software adaptable to new environment (both hardware and software)**
iv.	**Preventive**	**D.**	**Concerned with the change in the software that takes place to make the software adaptable to changing user requirements.**

Code :

	(i)	(ii)	(iii)	(iv)
(a)	(B)	(D)	(C)	(A)
(b)	(B)	(C)	(D)	(A)
(c)	(C)	(B)	(D)	(A)
(d)	(A)	(D)	(B)	(C)

Ans : (b)

(I) Corrective maintenance is concerned with fixing errors that are observed when the software is in use (b)

(II) Adaptive maintains is concerned with the change in the S/w that take place to the S/w adaptable to new environment (c)

(III) Perfective maintenance is concerned with implementing new or changed user requirements (c)

(IV) Preventive maintenance involves performing activities to prevent the occurrence of error (a)

45. Match each application/software design concept in List-I to its definition in List-II.

List-I		List-II	
i.	**Coupling**	**A.**	**Easy to visually inspect the design of the software and understand its purpose**
ii.	**Cohesion**	**B.**	**Easy to add functionality to a software without having to redesign it**
iii.	**Scalable**	**C.**	**Focus of a code upon a single goal.**
iv.	**Readable**	**D.**	**Reliance of a code module upon other code modules.**

Code :

	(i)	(ii)	(iii)	(iv)
(a)	(B)	(A)	(D)	(C)
(b)	(C)	(D)	(A)	(B)
(c)	(D)	(C)	(B)	(A)
(d)	(D)	(A)	(C)	(B)

Ans : (c)

46. Software safety is quality assurance activity that focuses on hazards that

(a) affect the reliability of a software component
(b) may cause an entire system to fail
(c) may result from user input errors
(d) prevent profitable marketing of the final product

Ans : (b) Software safety is a software quality assurance activity that focuses on the identification and assessment to potential hazards that may affect software negatively and cause an entire system to fail. If hazards can be identified early in the software process, software design factures can be specified that will either eliminate or control potential hazards.

47. Which of the following sets represent five stages defined by Capability Maturity Model (CMM) in increasing order of maturity?

(a) Initial, Defined, Repeatable, Managed, Optimized.
(b) Initial, Repeatable, Defined, Managed, Optimized.
(c) Initial, Defined, Managed, Repeatable, Optimized.
(d) Initial, Repeatable, Managed, Defined, Optimized.

Ans : (b) There are five levels defined along the continuum of the model and according to the SEI.

(1) Initial : The starting point for use of a new repeat process
(2) Repeatable : The process is at least documented sufficiently such that repeating the same steps may be attempted.
(3) Defined : The process is defined as a standard business process.
(4) Managed : This process is quantitatively managed in accordance with agreed upon metrics
(5) Optimizing : Process management includes deliberate process improvement.

48. The number of function points of a proposed system is calculated as 500. Suppose that the system is planned to be developed in Java and the LOC/FP ratio of Java is 50. Estimate the effort (E) required to complete the project using the effort formula of basic COCOMO given below :

$E = a(KLOC)^b$

Assume that the values of a and b are 2.5 and 1.0 respectively.

(a) 25 person months
(b) 75 person months
(c) 62.5 person months
(d) 72.5 person months

Ans : (c) E = a (KLOC)b
a = 2.5, b = 1.0
LOC| Fp = 50 ⇒10 c = 25000 i.e. 25 KLOC
$E = 2.5\ (25)^1 = 62.5$

49. In UNIX, processes that have finished execution but have not yet had their status collected are known as ....

(a) Sleeping processes (b) Stopped processes
(c) Zombie processes (d) Orphan processes

Ans : (c) **Zombie Process :** The UNIX system assumes that if child process terminate, the parent collects some data about the child, As soon as this collection is made, the system releases information bits about the child process and removes its pid from the process table. The UNIX is forced to keep the child's pid and termination data in the process table indefinitely. Such a terminated process whose status is not collected called a zombie process.

50. In Unix operating system, when a process creates a new process using the fork () system call, which of the following state is shared between the parent process and child process?

(a) Heap
(b) Stack
(c) Shared memory segments
(d) Both Heap and Stack

Ans : (c) Parent and child share only the share memory segments everything else (like stack, heap etc) is duplicated.

51. Which of the following information about the UNIX file system is not correct?

(a) Super block contains the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks.
(b) An i-node contains accounting information as well as enough information to locate all the disk block that holds the file's data.
(c) Each i-node is 256-bytes long.
(d) All the files and directories are stored in data blocks.

Ans : (c) **UNIX file system:**
Super block contains the number of i-nodes, the number of disk blocks, and the start of the list of free disk blocks.**Correct**
An i-node contains accounting information as well as enough information to locate all the disk blocks that holds the file's data.**Correct**
All the files and directories are stored in data blocks.**Correct**
Each i-node is 256-bytes long.**Incorrect**.It is 128 byte long.
So, option (C) is correct.

52. Which of the following option with reference to UNIX operating system is not correct?

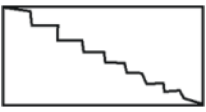
(a) INT signal is sent by the terminal driver when one types <Control> and it is a request to terminate the current operation.
(b) TERM is a request to terminate execution completely. The receiving process will clean up its state and exit.
(c) QUIT is similar to TERM, except that it defaults to producing a core dump if not caught.
(d) KILL is a block able signal.

Ans : (d) UNIX operating system does not kills a block able signal.

53. A multi computer with 256 CUPs is organized as 16 × 16 grid. What is the worst case delay (in hops) that a message might have to take?

(a) 16 (b) 15
(c) 32 (d) 30

Ans : (d)



In the 16 × 16 CPU case, the worst case delay happens when message passes through longest of the upper right corner to lower left corner or upper left corner to lower right corner.
Longest path is indicated in red color
It passes through 2(N –1) = 2(16 – 1) = 30 hops

54. Suppose that the time to do a null remote procedure call (RPC) (i.e. 0 data bytes) is 1.0 msec, with an additional 1.5 msec fro every 1K of data. How long does it take to read 32 K from the file server as 32 1K RPCs?

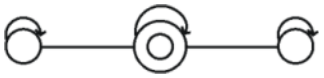
(a) 49 msec (b) 80 msec
(c) 48 msec (d) 100 msec

Ans : (b) A signal 32 K RPC takes 1.5 * 32 + 1.0 = 49.0 msec 321 K RPC take 1.5 * 32 + 1.0 * 32 = 80.0 msec

55. Let L be the language generated by regular expression 0*10* and accepted by the deterministic finite automata M. Consider the relation R_M defined by M. As all states are reachable from the start state, R_M has .... equivalence classes.

(a) 2 (b) 4
(c) 5 (d) 6

Ans : (d) 0* 10*



= equivalence class
= no. of states in DFA
= 3
Myhill nerorde equivalence relation = no. of states in minimal DFA

56. Let L = $\{0^n1^n | n \geq 0\}$ be a context free language. Which of the following is correct?

(a) $\overline{L}$ is context free and L^k is not context free for any $k \geq 1$.
(b) $\overline{L}$ is not context free and L^k is context free for any $k \geq 1$.
(c) Both $\overline{L}$ and L^k is for any $k \geq 1$ are context free.
(d) Both $\overline{L}$ and L^k is for any $k \geq 1$ are not context free.

Ans : (c) $S \rightarrow T| aU| vb$
$T \rightarrow aT| T| bT| Tb| ba$
$U \rightarrow aU| w$
$V \rightarrow Vb| w$
$W \rightarrow aWb|$
and L {0n 1n| n ≥ 03 is also context free (push 0 & pop 1 into stackand at last we have empty string) so both L1 & L2 are context free.

57. Given a Turing Machine

$M = (\{q_0, q_1, q_2, q_3\}, \{a, b\}, \{a, b, B\}\ \delta, B, \{q_3\})$

Where δ is a transition function defined as

$\delta(q_0, a) = (q_1, a, R)$

$\delta(q_1, b) = (q_2, b, R)$

$\delta(q_2, a) = (q_2, a, R)$

$\delta(q_2, b) = (q_3, b, R)$

The language L(M) accepted by the Turing Machine is given as :

(a) aa*b (b) abab

(c) aba*b (d) aba*

Ans : (c) Turing machine head movement only right direction we can build dfa for it.
Total 4 states are given
(Q_0, a) gives Q_1 no transition for b
(Q_1, b) gives Q_2 no transition for a
that means string with ab
now (Q_2, a) gives Q_2 that means it is loop on Q_2 from (Q_2, b) go to Q_3 and after that no transition that means string end with 'b'
So option (c) is answer.

58. Consider a discrete memory less channel and assume that H(x) is the amount of information per symbol at the input of the channel; H(y) is the amount of information per symbol at the output of the channel; H(x | y) is the amount of uncertainty remaining on x knowing y; and I(x; y) is the information transmission.

Which of the following does not define the channel capacity of a discrete memoryless channel?

(a) $\max_{p(x)} I(x ; y)$

(b) $\max_{p(x)} [H(y) - H(y|x)]$

(c) $\max_{p(x)} [H(x) - H(x|y)]$

(d) $\max_{p(x)} H(x|y)$

Ans : (d) The channel capacity of a discrete memory less channels is defined by :
C = max I (x; y) p (x)
= max [H(y) – H(y| x)] p(x)
in ax [H(x) – H(x)] p (x)

59. Consider a source with symbols A, B, C, D with probabilities 1/2, 1/4, 1/8, 1/8 respectively. What is the average number of bits per symbol for the Huffman code generated from above information?

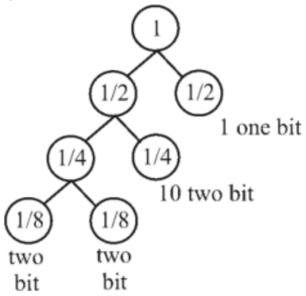
(a) 2 bits per symbol

(b) 1.75 bits per symbol

(c) 1.50 bits per symbol

(d) 1.25 bits per symbol

Ans : (b)



Length of bits 11

$$\frac{3\times1}{8}+\frac{3\times1}{8}+\frac{2\times1}{4}\times\frac{1\times1}{2}=\frac{14}{8}=1.75 \text{ bits per symbol}$$

60. Which of the following is used for the boundary representation of an image object?

(a) Quad tree (b) Projections
(c) Run length coding (d) Chain codes

Ans : (d) A chain is a lossless compression algorithm for monochrome image. The principle of chain nodes is separately encode each connected component in the image for each such regional a point on the boundary is selected and its coordinate are transmitted.

61. The region of feasible solution of a linear programming problem has a .... property in geometry, provided the feasible solution of the problem exists.

(a) concavity (b) convexity
(c) quadratic (d) polyhedron

Ans : (b) Geometric interpretation : The set of point in IRN at which any single constraint holds with equality is a hyper plane in IRN. Thus each constraint is satisfied by the points of a closed half space of IRN, & the set of feasible solution is the intersection of all these half spaces. Convex polyhedron p.
So answer is (b)

62. Consider the following statements :

(1) Revised simplex method requires lesser computations than the simplex method.
(2) Revised simplex method automatically generates the inverse of the current basis matrix.
(3) Less number of entries are needed in each table of revised simplex method than usual simplex method.

Which of these statements are correct?

(a) (1) and (2) only (b) (1) and (3) only
(c) (2) and (3) only (d) (1), (2) and (c)

Ans : (d) Revised simplex method is an improvement over simplex method. Revised simplex method is computational more efficiently and accurate. Benefit of revised simplex method is comprehended in case of large Lp problems. In simplex method tableau is updated while a small part of it is used. The revised simplex method use exactly the same steps as those in simplex method.
So answer is option (d)

63. **The following transportation problem :**

	A	B	C	Supply
I	**50**	**30**	**220**	**1**
II	**90**	**45**	**170**	**3**
III	**250**	**200**	**50**	**4**
Demand	**4**	**2**	**2**	

has a solution

	A	B	C
I	1		
II	3	0	
III		2	2

The above solution of a given transportation problem is :

(a) infeasible solution

(b) optimum solution

(c) non-optimum solution

(d) unbounded solution

Ans : (b)

(1)

$$T = \begin{array}{c} \\ X1 \\ X2 \end{array} \begin{array}{c} \begin{array}{ccc} Z1 & Z2 & Z3 \end{array} \\ \begin{bmatrix} 0.5 & 0.7 & 0.5 \\ 0.8 & 0.8 & 0.8 \end{bmatrix} \end{array}$$

(2)

$$T = \begin{array}{c} \\ X1 \\ X2 \end{array} \begin{array}{c} \begin{array}{ccc} Z1 & Z2 & Z3 \end{array} \\ \begin{bmatrix} 0.5 & 0.7 & 0.5 \\ 0.9 & 0.6 & 0.5 \end{bmatrix} \end{array}$$

(3)

$$T = \begin{array}{c} \\ X1 \\ X2 \end{array} \begin{array}{c} \begin{array}{ccc} Z1 & Z2 & Z3 \end{array} \\ \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix} \end{array}$$

(4)

$$T = \begin{array}{c} \\ X1 \\ X2 \end{array} \begin{array}{c} \begin{array}{ccc} Z1 & Z2 & Z3 \end{array} \\ \begin{bmatrix} 0.7 & 0.6 & 0.5 \\ 0.8 & 0.8 & 0.8 \end{bmatrix} \end{array}$$

64. Let R and S be two fuzzy relations defined as :

$$R = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} y_1 & y_2 \\ 0.7 & 0.5 \\ 0.8 & 0.4 \end{bmatrix}$$

and $$S = \begin{array}{c} \\ y_1 \\ y_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ 0.9 & 0.6 & 0.2 \\ 0.1 & 0.7 & 0.5 \end{bmatrix}$$

Then, the resulting relation, T, which relates elements of universe x to elements of universe z using max-min composition is given by

(a) $$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ .5 & .7 & .5 \\ .8 & .8 & .8 \end{bmatrix}$$

(b) $$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ .5 & .7 & .5 \\ .9 & .6 & .5 \end{bmatrix}$$

(c) $$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ 0.7 & 0.6 & 0.5 \\ 0.8 & 0.6 & 0.4 \end{bmatrix}$$

(d) $$T = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ 0.7 & 0.6 & 0.5 \\ 0.8 & 0.8 & 0.8 \end{bmatrix}$$

Ans : (c) $R = \begin{array}{c} \\ x_1 \\ x_2 \end{array} \begin{bmatrix} y_1 & y_2 \\ 0.7 & 0.5 \\ 0.5 & 0.4 \end{bmatrix}$ $S = \begin{array}{c} \\ y_1 \\ y_2 \end{array} \begin{bmatrix} z_1 & z_2 & z_3 \\ .9 & -6 & -5 \\ .8 & .6 & .4 \end{bmatrix}$

Then ROS by max min composition gives,

MROS (x_1, x_2) = max (min (.7, .9), min (.5, .1) = max (0.7, .1 = 0.7

as x_1 can be converted to z_1 through y_1 & y_2

MRos (x_1, z_2) = max (min (.7, .6), min (.5, .7) = max (.6, .5) = .6

similarly for all (x_1, z_3), (x_2, z_1) (x_2, z_2) (x_2, z_3)

So answer is (c)

65. Compute the value of adding the following two fuzzy integers :

A = {(0.3, 1), (0.6, 2), (1, 3), (0.7, 4), (0.2, 5)}

B = {(0.5, 11), (1, 12), (0.5, 13)}

Where fuzzy addition is defined as

$$\mu_{A+B}(z) = \max_{x+y=z} (\min(\mu_A(x), \mu_B(x)))$$

Then, f (A + B) is equal to

(a) {(0.5, 12), (0.6, 13), (1, 14), (0.7, 15), (0.7, 16), (1, 17), (1, 18)}

(b) {(0.5, 12), (0.6, 13), (1, 14), (1, 15), (1, 17), (1, 18)}

(c) {(0.3, 12), (0.5, 13), (0.5, 14), (1, 15), (0.7, 16), (0.5, 17), (0.2, 18)}

(d) {0.3, 12), (0.5, 13), (0.6, 14), 1, 15), (0.7, 16), (0.5, 17), 0.2, 18)}

Ans : (d) u A+ b (z) = max x + y = z (min u B (x)))
A = {0.3, 1), (0.6, 2), (1, 3), (0.7, 4), (0.2, 5)}
B = {(0.5, 11),)1, 12), (0.5, 13)}
First add the numbers (x + y = z) and write the main membership value since function is min ((uA (x), u B (x))
you will get following 15 terms
{0.3, 12), (0.3) 13), (0.3, 14), (0.5, 13), (0.6, 14), (0.5, 15), (0.5, 14), (1, 15), (0.5, 16), (0.5, 15), (0.7, 16), (0.5, 17), (0.2, 16), (0.2, 17), (0.2, 18)}
now write all distinct elements with max membership value
Since max is there in the question
So answer is (d)

66. A perceptron has input weights W1 = – 3.9 and W2 = 1.1 with threshold value T = 0.3. What output does it give for the input x1 = 1.3 and x2 = 2.2 ?

(a) – 2.65 (b) – 2.30
(c) 0 (d) 1

Ans : (c) According to given question $w_1 = -3.9$ and $w_2 = 1.1$ and $x_1 = 1.3$ and $x_2 = 2.2$ weighted sum = $w_1 * x_1 + w_2 * x_2 + \ldots w_n * x_n$ i.e. -3.9 * 1.3 + 1.1 * 2.2 = -5.07 + 2.42 = -2.65 Now we will compare the weighted sum -2.65 to the threshold 0.3. -2.65 < 0.3 Then output will be zero.
So, option (C) is correct.

67. What is the function of the following UNIX command?
WC – /<a > b&

(a) It runs the word count program to count the number of lines in its input, a, writing the result to b, as a foreground process.
(b) It runs the word count program to count the number of lines in its input, a, writing the result to b, but does it in the background.
(c) It counts the errors during the execution of a process, a, and puts the result in process b.
(d) It copies the '*l*' numbers of lines of program from file, a, and stores in file b.

Ans : (b) WC – 1<a > b&
wc for word cound-1 for line < a input a> b output b & for background
It runs the word count program to count the number of lines its input, a write result of b, but does it in the background.

68. Which of the following statement is not correct with reference to cron daemon in UNIX O.S.?

(a) The cron daemon is the standard tool for running commands on a pre-determined schedule.
(b) It starts when the system boots and runs as long as the system is up.
(c) Cron reads configuration files that contain list of command lines and the times at which they invoked.
(d) Cronatab for individual users are not stored.

Ans : (d) Cron daemon is the standard tool for running command an a receteminced schedule. It starts when the system boots and run as long as the system is up. Crontabs for individual u>a are stored under var| spool| cron. There is one crontab file per user.

69. In Unix, files can be protected by assigning each one a 9-bit mode called right bits. Now, consider the following two statements :

I. A mode of 641 (octal) means that the owner can read and write the file, other members of the owner's group can read it, and users can execute only.

II. A mode of 100 (octal) allows the owner to execute the file, but prohibits all other access.

Which of the following options is correct with reference to above statements?

(a) Only I is correct
(b) Only II is correct
(c) Both I and II are correct
(d) Both I and II are incorrect

Ans : (c) In UNIX the access control of file is done through chmod command. The common is octal, the there right most are for
First is for owner, second is for group, third is for other here rwx is read, write execute. This can take the value from 0 through 7 where

#	permission	rwx
7	read, write & execute	rwx
6	read, write	rw–
5	read & execute	r–x
4	read only	r– –
3	write & execute	–wx
2	write only	–w–
1	execute only	– –x
0	none	– –

641 represent → The owner can read and write the file, other member of the owner group can read it & user can execute only
100 → The owner to execute the file, but prohibits all other access

70. Consider the statement

"Either $-2 \le x \le -1$ or $1 \le x \le 2$"

The negation of this statement is

(a) $x < -2$ or $2 < x$ or $-1 < x < 1$
(b) $x < -2$ or $2 < x$
(c) $-1 < x < 1$
(d) $x \le -2$ or $2 \le x$ or $-1 < x < 1$

Ans : (a) "Either $-2 \le x \le -1$ or $1 \le x \le 2$". i.e. Either $x \ge -2$ or $x \le -1$ or $1 \le x$ or $x \le 2$" We have to find negation of above statement: Negation of $x \ge -2$ is $x < 2$. Negation of $x \le -1$ is $x > -1$. Negation of $1 \le x$ is $x < 1$ Negation of $x \le 2$ is $x > 2$. i.e. $x < -2$ or $2 < x$ or $-1 < x < 1$. So, option (A) is correct.

71. Which of the following is characteristic of an MIS?

(a) Provides guidance in identifying problems, finding and evaluating alternative solutions, and selecting or comparing alternatives.

(b) Draws on diverse yet predictable data resources to aggregate and summarize data.

(c) High volume, data capture focus.

(d) Has as its goal the efficiency of data movement and processing and interfacing different TPS.

Ans : (b) Choice - (1) refers to characteristics of DSS (Decision support system)
Choice (3, 4) refers to characteristic for a transaction processing system, so answer is (b)

72. How does randomized hill-climbing choose the next move each time?

(a) It generates a random move from the move set, and accepts this move.

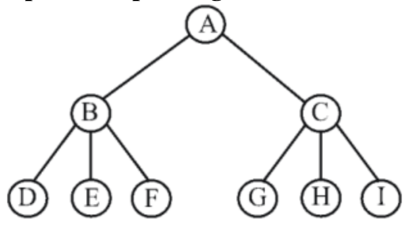
(b) It generates a random move from the whole state space, and accepts this move.

(c) It generates a random move from the moveset, and accepts this move only if this move improves the evaluation function.

(d) It generates a random move from the whole state space, and accepts this move only if this move improves the evaluation function.

Ans : (c) **Randomized Hill climbing :** Sample D points in the neighborhood of the con currently best solution; determine the best solution of the sampled points, It is better than the current solution, make it new current solution & continue the search.

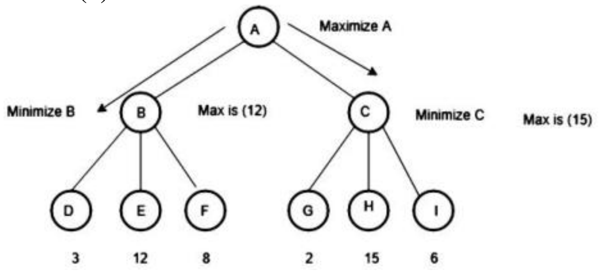
73. Consider the following game tree in which root is a maximizing node and children are visited left to right. What nodes will be pruned by the alpha-beta pruning?



(a) I (b) HI

(c) CHI (d) GHI

Ans : (b)



We have to maximize A and minimize B and C. Minimum of B is 3 and C is 2 But max from B is 12 and max from C is 15. So, there is no need of further expanding H and I. Because whatever value their successor will produce will be immaterial. Hence H I are pruned. For more information on Game theory Refer : Minimax Algorithm in Game Theory | Set 4 (Alpha-Beta Pruning) Option (B) is correct.

74. Consider a 3-puzzle where, like in the usual 8-puzzle game, a tile can only move to an adjacent empty space. Given the initial state

1	2
	3

, which of the following state cannot be reached?

(a)

3	1
	2

(b)

	3
2	1

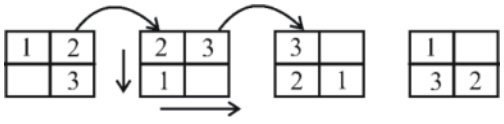
(c)

1	3
	2

(d)

	2
1	3

Ans : (c)



Not able to get (c)
So answer is option (c)

75. A software program that infers and manipulates existing knowledge in order to generate new knowledge is known as :

(a) Data dictionary
(b) Reference mechanism
(c) Inference engine
(d) Control strategy

Ans : (c) A software program that infers and manipulates existing knowledge in order to generate new knowledge is known as **Inference engine**.

A data dictionary is a collection of descriptions of the data objects or items in a data model for the benefit of programmers and others who need to refer to them.

Control strategies are specific action plans for bringing a process back into control.

Reference is what relates words to the world of objects on whose condition the truth of sentences hinges and it is natural to wonder what sorts of relations underlie the reference relation-to wonder, that is, what constitutes the mechanism of reference.

So, option (C) is correct.